Take $x = \tan\theta$.

Then $\dfrac{2x}{1-x^2} = \dfrac{2\tan\theta}{1-\tan^2\theta} = \tan 2\theta$ and

$$f(x) = \tan^{-1}(\tan 2\theta) = 2\theta$$
$$= 2\tan^{-1}x$$
$$f'(x) = \frac{2}{1+x^2}.$$

Example 10.24

If $y = \tan^{-1}\left(\dfrac{1+x}{1-x}\right)$, find y'.

Solution

Let $x = \tan\theta$

$$\text{Then } \frac{1+x}{1-x} = \frac{1+\tan\theta}{1-\tan\theta} = \tan\left(\frac{\pi}{4}+\theta\right).$$

$$\tan^{-1}\left(\frac{1+x}{1-x}\right) = \tan^{-1}\left[\tan\left(\frac{\pi}{4}+\theta\right)\right] = \frac{\pi}{4}+\theta = \frac{\pi}{4}+\tan^{-1}x$$

$$y = \frac{\pi}{4}+\tan^{-1}x$$

$$y' = \frac{1}{1+x^2}.$$

Example 10.25

Find $f'(x)$ if $f(x) = \cos^{-1}(4x^3-3x)$.

Solution

Let $x = \cos\theta$.

$$\text{Then } 4x^3-3x = 4\cos^3\theta - 3\cos\theta = \cos 3\theta \text{ and}$$

$$f(x) = \cos^{-1}(\cos 3\theta) = 3\theta = 3\cos^{-1}x$$

$$\text{Therefore, } f'(x) = 3\left(\frac{-1}{\sqrt{1-x^2}}\right) = \frac{-3}{\sqrt{1-x^2}}.$$

10.4.6 Derivatives of variables defined by parametric equations

Consider the equations $x = f(t)$, $y = g(t)$.

These equations give a functional relationship between the variables x and y. Given the value of t in some domain $[a, b]$, we can find x and y.

If two variables x and y are defined separately as a function of an intermediating (auxiliary) variable t, then the specification of a functional relationship between x and y is described as **parametric** and the auxiliary variable is known as **parameter**.

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